

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10 – SUB-STRAND 2.9 LINEAR MOTION: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: How can we use mathematics, through calculations and graphs, to fully describe and explain the motion of any moving body, from a Nairobi Marathon runner to a matatu on the Nakuru highway? In the 2023 Nairobi Marathon, Runner A (nearly constant pace) won, even though Runner B (fast start, slow middle, sprint finish) was ahead at 30 km. Use everything you learned in Lessons 1 to 8 to answer the five sections below. Write in full sentences, show every step of your working, include units, and refer back to the marathon in your answers. Where a sketch is asked for, draw it neatly with labelled axes. Use your class notes, graphs and Driving Question Board to help you.

Section 1: The language of motion (Key Inquiry Question 1: What is speed?)

(a) Explain the difference between distance and displacement, and between speed and velocity. State which of these are scalar quantities and which are vector quantities. (b) Both runners covered 42 km in the Nairobi Marathon. Explain why comparing only their total times could not tell you that Runner B was ahead at 30 km. (c) Define acceleration and say which part of Runner B's race it helps us describe.

(a) Distance is the total length of the path covered. It is a scalar quantity because it has size only and no direction. Displacement is the straight-line change in position from start to finish, including direction, so it is a vector quantity. Speed = distance ÷ time and is a scalar. Velocity = displacement ÷ time with direction and is a vector. (b) A total time and a single average speed only use the start and end of the race. They hide what happens along the way. Runner B ran fast early, slowed in the middle and sprinted at the end, while Runner A ran at a nearly constant pace of about 19 km/h. A single average cannot show that B led at 30 km but then lost ground, so we need velocities for each section and graphs. (c) Acceleration is the rate of change of velocity, $a = \Delta v / \Delta t$, measured in m/s^2 . It describes Runner B speeding up off the start (positive acceleration), slowing down in the middle (negative acceleration, or deceleration) and speeding up again in the final sprint.

Section 2: Calculating velocity and acceleration

(a) Runner A ran at a nearly constant velocity of about 19 km/h. Use $v = s/t$ to find how far Runner A runs in 2 hours. (b) Runner B's velocity in the middle of the race was about 17.6 km/h. Calculate how much

(a) $s = v \times t = 19 \text{ km/h} \times 2 \text{ h} = 38 \text{ km}$. (b) Runner A: $19 \times 0.5 = 9.5 \text{ km}$. Runner B: $17.6 \times 0.5 = 8.8 \text{ km}$. Difference = $9.5 - 8.8 = 0.7 \text{ km}$, so A runs 0.7 km (700 m) further in 30 minutes. (c) $a = \Delta v / \Delta t$, so $\Delta v = a \times \Delta t = 1 \times 10 = 10 \text{ m/s}$. Starting from rest, the velocity after 10 s is 10 m/s. (d) Runner A's motion is uniform because his velocity is nearly constant, so his acceleration is about zero. A negative acceleration (deceleration) means the body is slowing down, which is what Runner B did in the middle of the race.

further Runner A runs than Runner B in 30 minutes (0.5 h) at these velocities. (c) A boda-boda on Mombasa Road accelerates from rest at 1 m/s^2. Use $a = \Delta v/\Delta t$ to find its velocity after 10 seconds. (d) Is Runner A's motion uniform or non-uniform? What is his acceleration? What does a negative acceleration mean for Runner B?	
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Section 3: Displacement-time graphs (Key Inquiry Question 2: How do we use displacement-time graphs in real life?)

(a) Say what each of these shapes on a displacement-time graph tells you about the motion: a steep straight line, a gentle straight line, a horizontal line, a line sloping back down. (b) State what the gradient of a displacement-time graph measures. (c) Describe in words how the lines for Runner A and Runner B would look on the same displacement-time graph, from the start to the finish. Say where the lines cross and what the crossing shows. (d) Give one real-life Kenyan example where a displacement-time graph is useful, for example a matatu on the Nakuru highway.	(a) A steep straight line means high velocity. A gentle straight line means low velocity. A horizontal line means zero gradient, so the body is stationary. A line sloping back down means a negative gradient, so the body is returning towards its starting point. (b) The gradient is the velocity (change in displacement $\div$ change in time). (c) Runner A's line is a nearly straight line with a steady gradient of about 19 km/h. Runner B's line starts steeper, becomes noticeably less steep in the middle section (he slows to about 17.6 km/h), and steepens again at the end (the sprint). Runner B's line is above A's at 30 km, but because A's line keeps its steady gradient, it crosses B's line after 30 km, in the slow section. The crossing point is when Runner A overtakes Runner B. After it, A stays ahead and wins. (d) For a matatu from Nakuru town, a displacement-time graph shows a steep line while it is moving, a horizontal line while it stops at a stage, and the gradient of each section gives its velocity. This tells us about each part of the journey, not just the overall average.
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Section 4: Velocity-time graphs (Key Inquiry Question 3: How do we use velocity-time graphs?)

(a) State what the gradient of a velocity-time graph measures. State what the area between the graph and the time axis measures. (b) Sketch a velocity-time graph for a matatu on the	(a) The gradient of a velocity-time graph is acceleration ($a = \Delta v/\Delta t$). The area between the graph and the time axis is the displacement (distance travelled in a straight line). (b) The sketch has velocity on the vertical axis (m/s or km/h) and time on the horizontal axis. It shows a rising straight line labelled acceleration (positive gradient), then a horizontal line labelled constant velocity (zero acceleration), then a falling straight line labelled deceleration (negative gradient) down towards zero. (c) The area is a rectangle, so area = velocity $\times$ time. Runner A: $19 \text{ km/h} \times 0.5 \text{ h} = 9.5 \text{ km}$. Runner B: $17.6 \text{ km/h} \times 0.5 \text{ h} = 8.8 \text{ km}$. Runner A's area is larger by 0.7 km. In this slow section Runner A covered more ground than Runner B, which is why Runner B lost the lead he had
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Nakuru–Nairobi highway that accelerates from a stage, cruises at constant velocity, then brakes. Label each of the three sections as acceleration, constant velocity or deceleration. (c) For the 30-minute part of the race in Section 2(b), the velocity-time graph for each runner is a horizontal line. Use area under the graph to find the distance each runner covers, and say whose area is larger and what that means for the race.	at 30 km.
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Section 5: Relative speed and the full story of the marathon	
(a) Runner B had slowed to about 17.6 km/h and Runner A was running at about 19 km/h. Calculate the relative speed of Runner A with respect to Runner B. Then find how many minutes Runner A needs to close a gap of 200 m (0.2 km) to Runner B. (b) A matatu travelling at 90 km/h overtakes a bus travelling at 60 km/h in the same direction. Find the relative speed. (c) Two vehicles each travelling at 80 km/h approach each other in opposite directions. Find the relative speed, and explain why this differs from part (b). (d) Now write a final explanation of the driving question. Use the Nairobi Marathon to explain why Runner A won even though Runner B was ahead at 30 km. In your answer refer to average speed and instantaneous velocity, acceleration, the shape of a displacement-time graph, and a velocity-time	(a) Both runners move in the same direction, so relative speed = faster – slower = $19 - 17.6 = 1.4$ km/h. Time = gap ÷ relative speed = $0.2 \div 1.4 = 0.143$ h, which is about 8.6 minutes (0.143×60). (b) Same direction: $90 - 60 = 30$ km/h. (c) Opposite directions: relative speed = $80 + 80 = 160$ km/h. When bodies move in opposite directions their speeds add, but in the same direction they are subtracted, so the closing speed is much greater (160 km/h compared with 30 km/h). (d) A single average speed or total time cannot show what happened at each point of the race, so we need velocities, acceleration and graphs. Runner A kept a nearly constant velocity of about 19 km/h, so his acceleration was about zero. Runner B accelerated off the start and was ahead at 30 km. In the middle section he decelerated, his velocity fell to about 17.6 km/h, and he covered less distance than Runner A. On the displacement-time graph B's line became less steep (smaller gradient = lower velocity), while A's stayed steady, so A's line crossed B's after 30 km, when A overtook him. On the velocity-time graph B's line fell (negative gradient) in the middle, so the area under his graph, which is his distance, was smaller than A's for that section. His final sprint was a positive acceleration, but it came too late to recover the lost time. Because A only needed to be faster than B's current speed (relative speed of about 1.4 km/h), he gradually closed the gap and won. Mathematics lets us describe and explain motion exactly: the shape of the graph, not just the start and end points, tells the full story.

graph.	
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of key terms (distance/displacement, speed/velocity, scalar/vector, acceleration)	Defines all terms correctly, correctly labels each as scalar or vector, and uses them accurately and consistently throughout.	Defines most terms correctly with only minor slips in labelling scalar or vector.	Confuses distance with displacement or speed with velocity, or omits scalar/vector.
Calculations ($v = s/t$, $a = \Delta v/\Delta t$, area under graph, relative speed)	All calculations are correct with clear working, correct formulas and correct units.	Most calculations are correct; occasional errors in units or working.	Several incorrect calculations or formulas, little working or missing units.
Graph interpretation (gradient and area)	Correctly states that the gradient of a displacement-time graph is velocity, the gradient of a velocity-time graph is acceleration and the area is displacement. Sketches are neat, labelled, and linked to Runner A and Runner B.	States the meaning of gradient and area correctly, but sketches or links to the runners are partly incomplete.	Mixes up what the gradient or area means on the two types of graph, or sketches are unlabelled.
Explaining the phenomenon with evidence	Gives a complete, logical explanation of why Runner A won despite Runner B leading at 30 km, using velocity, acceleration, graph shape and relative speed.	Explains the main reason (B slowed in the middle) using at least two ideas from the unit.	Only restates that A won, or uses average speed alone, with little evidence.
Kenyan contexts and communication	Links ideas clearly to Kenyan examples (matatu, boda-boda, Nairobi Marathon) in clear, well-organised sentences.	Uses some Kenyan examples; writing is mostly clear.	Few or no links to the contexts; explanations are unclear or very short.

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